ModelIt!

Student Activity Pack

The Concussion Problem

Why Helmets Can't Stop Your Brain from Slamming Into Your Skull

Grade 8 | NGSS MS-PS2-1, MS-PS2-2 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens to a passenger when a car suddenly stops?

2 Why do you think football players get concussions even with helmets?

3 What does Newton's Third Law say about collisions?

4 Draw what you think happens to the brain inside a skull during a sudden stop.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Collision Speed | Player Mass | Impact Force | Brain Acceleration

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Jogging Speed

5 mph collision, normal mass

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Full Sprint

Lock Collision Speed to 20 mph

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Heavier Player

Double Player Mass at 15 mph

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Padding Thickness — Thicker padding extends the time of impact, reducing peak force

Neck Strength — Stronger neck muscles help stabilize the head during impact

Impact Angle — Glancing blows transfer less force than head-on collisions

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do football players still get concussions even while wearing helmets? Use evidence from your simulation to support your answers.

1. **Why can't any helmet completely prevent concussions?**

2. **How does Newton's First Law explain what happens to the brain during a collision?**

3. **Why is a 20 mph collision so much more dangerous than a 10 mph collision?**

4. **How is an egg drop contest similar to designing a better helmet?**

5. **Should contact sports be modified to reduce collision speeds? What are the trade-offs?**

8 STEM Challenge: Design a Better Helmet

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YOUR ENGINEERING MISSION

Design a helmet that minimizes brain acceleration during impacts using the principles from your model.

THE SCENARIO

A sports equipment company has hired your team to improve their football helmet design. You must use physics principles to explain why your design reduces concussion risk.

Engineering Design Thinking

1. WHAT PADDING MATERIALS AND THICKNESS WILL YOU USE?
2. HOW WILL YOUR DESIGN EXTEND THE TIME OF IMPACT?
3. WHAT'S THE TRADE-OFF BETWEEN PROTECTION AND WEIGHT/COMFORT?
4. HOW WILL YOU TEST YOUR HELMET DESIGN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomedical Engineers design safety equipment using physics and biology. Sports Biomechanists study forces on the human body during athletics. They earn $70,000-$120,000/year.